

1. Executive Summary

Metric	Value	Traffic Summary	Count
Report Period	2026-08-31 - 2026-09-07	Total Flows Observed	14171
Total Threats Mitigated	32	Total Packets Analyzed	1970225561
High Priority	30	Malicious Packets	1969106756
Low Priority	2	True Negatives Passed	6537
Manual Releases	0		
Manual Blocks	0		
ML Metrics	Value	Attack Vector	Count
False Positives	442	ICMP Flood	6
FP Rate	6.33%	SYN Flood	12
IF Precision	0.00%	UDP Flood	12
IF Recall	0.00%	Uncertain	2
IF F1-Score	0.00%		
IF Accuracy	93.67%	Action	Count
		Quarantined	1
		Time Ban	9
		Blackhole	0
		Released	3

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	0.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	0.00%	Share of actual attacks successfully flagged
F1-Score	0.00%	Balanced measure combining Precision and Recall
Accuracy	93.67%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	6.33%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	0.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	93.67%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 0	FN 0
Actual: Normal	FP 442	TN 6537

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	0.00%	Share of classified flows assigned the correct attack type
Recall	0.00%	Share of attacks of each type that were correctly identified
F1-Score	0.00%	Balanced measure combining Precision and Recall
Accuracy	0.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Response Latency

Metric	Value	Description
Detection Time	462.1 ms	Average time taken to detect an attack
Mitigation Response Time	2.8 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.00%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.22%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.22%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.10	2	2	2026-09-07 13:56:52	2026-09-07 13:58:44
10.0.0.16	1	2	2026-09-07 13:56:43	2026-09-07 13:57:44
10.0.0.17	1	2	2026-09-07 13:56:43	2026-09-07 13:57:44
10.0.0.18	1	2	2026-09-07 13:56:43	2026-09-07 13:57:44
10.0.0.19	1	2	2026-09-07 13:56:43	2026-09-07 13:57:44
10.0.0.20	1	2	2026-09-07 13:56:43	2026-09-07 13:57:44
10.0.0.21	1	2	2026-09-07 13:56:43	2026-09-07 13:57:44
10.0.0.22	1	2	2026-09-07 13:56:43	2026-09-07 13:57:44
10.0.0.23	1	2	2026-09-07 13:56:44	2026-09-07 13:57:45
10.0.0.25	1	2	2026-09-07 13:56:44	2026-09-07 13:57:44
10.0.0.2	9	1	2026-09-07 13:56:44	2026-09-07 13:58:47
10.0.0.3	1	1	2026-09-07 13:56:45	2026-09-07 13:56:56

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-07 13:56:43	10.0.0.21	DDoS	UDP Flood	75.6%	High	Time Ban (1m)
2026-09-07 13:56:43	10.0.0.22	DDoS	UDP Flood	75.6%	High	Time Ban (1m)
2026-09-07 13:56:43	10.0.0.21	DDoS	UDP Flood	75.6%	High	Time Ban
2026-09-07 13:56:43	10.0.0.16	DDoS	SYN Flood	90.4%	High	Time Ban (1m)
2026-09-07 13:56:43	10.0.0.22	DDoS	UDP Flood	75.6%	High	Time Ban
2026-09-07 13:56:43	10.0.0.17	DDoS	SYN Flood	89.4%	High	Time Ban (1m)
2026-09-07 13:56:43	10.0.0.19	DDoS	SYN Flood	90.4%	High	Time Ban (1m)
2026-09-07 13:56:43	10.0.0.20	DDoS	UDP Flood	76.8%	High	Time Ban (1m)
2026-09-07 13:56:43	10.0.0.18	DDoS	SYN Flood	89.2%	High	Time Ban (1m)
2026-09-07 13:56:43	10.0.0.16	DDoS	SYN Flood	90.4%	High	Time Ban
2026-09-07 13:56:43	10.0.0.17	DDoS	SYN Flood	89.4%	High	Time Ban
2026-09-07 13:56:43	10.0.0.19	DDoS	SYN Flood	90.4%	High	Time Ban
2026-09-07 13:56:43	10.0.0.20	DDoS	UDP Flood	76.8%	High	Time Ban
2026-09-07 13:56:43	10.0.0.18	DDoS	SYN Flood	89.2%	High	Time Ban
2026-09-07 13:56:44	10.0.0.25	DDoS	ICMP Flood	92.9%	High	Time Ban (1m)
2026-09-07 13:56:44	10.0.0.25	DDoS	ICMP Flood	92.9%	High	Time Ban
2026-09-07 13:56:44	10.0.0.23	DDoS	ICMP Flood	93.1%	High	Time Ban (1m)
2026-09-07 13:56:44	10.0.0.23	DDoS	ICMP Flood	93.1%	High	Time Ban
2026-09-07 13:56:52	10.0.0.10	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-07 13:56:55	10.0.0.2	DDoS	Uncertain	59.7%	Low	Released
2026-09-07 13:56:56	10.0.0.3	DDoS	Uncertain	58.4%	Low	Released
2026-09-07 13:57:44	10.0.0.21	DDoS	UDP Flood	79.5%	High	Time Ban (2m)
2026-09-07 13:57:44	10.0.0.22	DDoS	UDP Flood	82.3%	High	Time Ban (2m)
2026-09-07 13:57:44	10.0.0.16	DDoS	SYN Flood	90.4%	High	Time Ban (2m)
2026-09-07 13:57:44	10.0.0.17	DDoS	SYN Flood	89.4%	High	Time Ban (2m)
2026-09-07 13:57:44	10.0.0.19	DDoS	SYN Flood	90.4%	High	Time Ban (2m)
2026-09-07 13:57:44	10.0.0.20	DDoS	UDP Flood	77.6%	High	Time Ban (2m)
2026-09-07 13:57:44	10.0.0.18	DDoS	SYN Flood	89.7%	High	Time Ban (2m)
2026-09-07 13:57:44	10.0.0.25	DDoS	ICMP Flood	95.8%	High	Time Ban (2m)
2026-09-07 13:57:45	10.0.0.23	DDoS	ICMP Flood	93.1%	High	Time Ban (2m)
2026-09-07 13:57:52	10.0.0.10	DDoS	UDP Flood	92.4%	High	Released
2026-09-07 13:58:24	10.0.0.10	DDoS	UDP Flood	92.4%	High	Quarantined

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.2	Uncertain	0.6019	59.7%	Medium	Phase 1	1	15s	2026-09-07 13:58:47	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.10	UDP Flood	0.5799	92.4%	High	Phase 1	2	20s	2026-09-07 13:58:44	Attack Stopped
10.0.0.2	Uncertain	0.6156	59.7%	Medium	Phase 1	1	16s	2026-09-07 13:58:21	Attack Stopped
10.0.0.2	Uncertain	0.6166	59.7%	Medium	Phase 1	1	16s	2026-09-07 13:58:05	Attack Stopped
10.0.0.10	UDP Flood	0.5564	92.4%	High	Phase 2	1	1m 0s	2026-09-07 13:57:52	Ban Expired
10.0.0.2	Uncertain	0.6160	59.7%	Low	Phase 1	1	10s	2026-09-07 13:57:49	Attack Stopped
10.0.0.23	ICMP Flood	0.5883	93.1%	High	Phase 2	2	1m 1s	2026-09-07 13:57:45	Escalated to Time Ban
10.0.0.21	UDP Flood	0.6275	79.5%	High	Phase 2	2	1m 1s	2026-09-07 13:57:44	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6277	82.3%	High	Phase 2	2	1m 1s	2026-09-07 13:57:44	Escalated to Time Ban
10.0.0.16	SYN Flood	0.6122	90.4%	High	Phase 2	2	1m 1s	2026-09-07 13:57:44	Escalated to Time Ban
10.0.0.17	SYN Flood	0.6131	89.4%	High	Phase 2	2	1m 1s	2026-09-07 13:57:44	Escalated to Time Ban
10.0.0.19	SYN Flood	0.6161	90.4%	High	Phase 2	2	1m 1s	2026-09-07 13:57:44	Escalated to Time Ban
10.0.0.20	UDP Flood	0.6236	77.6%	High	Phase 2	2	1m 1s	2026-09-07 13:57:44	Escalated to Time Ban
10.0.0.18	SYN Flood	0.6147	89.7%	High	Phase 2	2	1m 1s	2026-09-07 13:57:44	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6375	95.8%	High	Phase 2	2	1m 0s	2026-09-07 13:57:44	Escalated to Time Ban
10.0.0.2	Uncertain	0.6114	59.7%	Low	Phase 1	1	11s	2026-09-07 13:57:39	Attack Stopped
10.0.0.2	Uncertain	0.6114	59.7%	Low	Phase 1	1	11s	2026-09-07 13:57:28	Attack Stopped
10.0.0.2	Uncertain	0.6114	59.7%	Low	Phase 1	1	11s	2026-09-07 13:57:17	Attack Stopped
10.0.0.2	Uncertain	0.6172	59.7%	Low	Phase 1	1	11s	2026-09-07 13:57:06	Attack Stopped
10.0.0.3	Uncertain	0.5924	58.4%	Low	Phase 1	1	11s	2026-09-07 13:56:56	Attack Stopped
10.0.0.2	Uncertain	0.6128	59.7%	Low	Phase 1	1	11s	2026-09-07 13:56:55	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

DATE

Network Administrator